Six math clues · answer key

Tiny Guests

Age 8 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

Math 1 · 45

28 + 17 = 45. Add 10 to reach 38, then add 7 to reach 45.

Math 2 · 12

20 − 8 = 12. After 8 seconds, 12 seconds remain.

Math 3 · 42

24 + 18 = 42. First reach 34, then add 8 to reach 42.

Math 4 · 36

52 − 16 = 36. Subtract 10 to reach 42, then subtract 6 to reach 36.

Math 5 · 62

35 + 27 = 62. Add 20 to reach 55, then add 7 to reach 62.

Math 6 · 38

63 − 25 = 38. Subtract 20 to reach 43, then subtract 5 to reach 38.

Age 8 / Grade 2 math. Six one-step story problems within 100, with three additions and three subtractions. Offer counters, tens-and-ones drawings, or spoken reasoning.

Reading & writing · answer key

Tiny Guests

Reading · 1

A bacterium is a whole living cell. A virus is smaller and is not a cell; it carries instructions in a coat and needs a living cell’s machinery to make copies.

Reading · 2

Soap can pull apart oily layers of many germs and helps loosen them from skin. Scrubbing helps lift them, and running water carries them away.

Reading · 3

Some defenders remember a germ’s features. Remembering can help them recognize the same germ faster next time.

Writing

Accept one sensible command, spoken, dictated, or written. Example: “First wet your hands, then scrub with soap for 20 seconds.”

Age 8: Grade 3 reading and Grade 2 math. Grown-ups and doctors decide about medicine. Many bacteria are helpful. Soap disrupts oily membranes of susceptible germs; it does not break apart every bacterium or every virus. Scrubbing and rinsing remove germs too. The diorama compares a bacterium and a virus with an oily outer layer; its removal amounts are qualitative, not measured. Antibiotics differ in which bacteria they affect and do not act on viruses. Immune memory helps recognition but is not a promise of complete protection.

Accept a spoken command, dictation, invented spelling, or writing. One clear sentence is enough. Example: “First wet your hands, then scrub with soap for 20 seconds.”

Four clues to keep

Tiny Guests

Quiz 1. Which tiny guest is a whole living cell?

A bacterium is a whole living cell. A virus is a package of instructions in a coat and needs a living cell to make copies.

Quiz 2. How do soap, scrubbing, and rinsing work together?

Soap can pull apart oily outer layers of many germs. Soap and scrubbing also lift germs off skin so water can rinse them away. Scrub for about 20 seconds.

Quiz 3. Which tiny guests can antibiotics work on?

Antibiotics work on bacteria, but not on viruses. Viruses do not have the same cell parts. Grown-ups and doctors decide about medicine.

Quiz 4. What can some of your body’s defenders do?

Some defenders in your immune system learn to recognize germs and remember their features. That can help them recognize the same germs faster next time.

Age 8: Grade 3 reading and Grade 2 math. Grown-ups and doctors decide about medicine. Many bacteria are helpful. Soap disrupts oily membranes of susceptible germs; it does not break apart every bacterium or every virus. Scrubbing and rinsing remove germs too. The diorama compares a bacterium and a virus with an oily outer layer; its removal amounts are qualitative, not measured. Antibiotics differ in which bacteria they affect and do not act on viruses. Immune memory helps recognition but is not a promise of complete protection.